

CMPE2150 PCB Project C

AJ Armstrong

09 Dec 2024

Table of contents

PCB Layout 1
Part 1 - Setting Up the Board 1
Part 2 - Component Placement 2



Tip

I'm going to stop reminding you to increment version numbers, check in and push at major milestones, because you already do that reflexively.

PCB Layout

Now that we have a fully annotated and checked schematic, we can begin the process of layout out a PCB. This process will follow the procedure in the tutorial rather closely, so most of the tasks should be reasonably familiar. This is always true, but particularly in PCB layout it is important to be neat, to check things carefully and to pay attention to constraints and instructions. Very subtle errors on a PCB design can make the boards useless after manufacture and waste hundreds of dollars and weeks of time on even a small test run. Your instructors will rarely have time to do more than a cursory check of this work, but you are encouraged to review each others' work and offer feedback and suggestions, up to the normal limits of not sharing designs.

Part 1 - Setting Up the Board

Open your project's PCB editor and:

- Select *File* → *Page Settings* and set the page up the same way we did for the schematic in Part A.

- Change the default units from **mm** to **mils** by selecting the *mil* button on the left toolbar. Mils¹ are far more commonly used in North America.
- On the top menu bar, change the grid spacing to 20 **mils**.

¹ Thousandths of an inch

Now, go to *File* → *Board Setup...* and make or confirm the following settings:

- *Board Stackup* → *Physical Stackup*: Leave as 2 copper layers, as this is still a relatively simple board.
- *Text & Graphics* → *Defaults*: Change *Units* to **mils** then round the default properties at the top up to the nearest mil.
- *Design Rules* → *Design Constraints*: Round constraint values up to the nearest whole ‘ mil
- *Design Rules* → *Net Classes*: Round constraint values up to the nearest whole mil.
- You may leave the remaining settings at default values.

Part 2 - Component Placement

Submission

When you have completed all the above—including having no ERC errors or warnings—and checked and reviewed your work, ask your instructor to review your schematic with you. Make any changes they require. When you are done submit the schematic.